RWA Tokenization Platform

Production-Grade Decentralized Protocol for Real-World Assets

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Arbitrum Sepolia L2 · May 2026 · BChT2 Final Project

Project Overview

What is RWA

Real-World Assets tokenised on-chain – bonds, real estate, commodities

Why Blockchain

Transparency · Immutability · Automated enforcement · Global access

Why Arbitrum L2

≈90% lower gas · near-instant finality · inherits Ethereum security

Our Solution

Full-stack protocol: issuance, trading, yield, governance for RWAs

Impact

Institutional access to RWAs with lower friction & cost

System Architecture

Asset Tokenization

AssetFactory → deterministic CREATE2 ERC-20 tokens. V1 (base) → UUPS upgradeable V2 (KYC gates)

Yield Vault

ERC-4626 vault: shares, deposits, automated yield distribution

Decentralized Exchange

RWAAMM (x·y=k) with 0.3% fees + slippage protections

DAO Governance

Governor + 2-day Timelock for upgrades and parameter changes

Smart Contracts – Deployed (10)

- **AssetFactory**
  CREATE2 deterministic deployment · role-based access
- **AssetToken V1/V2**
  ERC-20 + UUPS proxies · storage safety for upgrades
- **AssetNFT**
  ERC-721 legal certificates per asset
- **RWAVault**
  ERC-4626 vault · ReentrancyGuard · fee hooks
- **RWAAMM**
  Custom constant-product AMM – built from scratch
- **ChainlinkOracleAdapter**
  Staleness checks · Yul gas optimisations
- **GovernanceToken**
  ERC20Votes + ERC20Permit (gasless voting)
- **RWAGovernor & RWATimelockController**
  OpenZeppelin governance stack for proposals & execution

Security & Design Patterns

CEI Pattern

Checks → Effects → Interactions on all external functions

ReentrancyGuard

Protects vault deposits & AMM swaps

AccessControl

Role-based permissions; no single god key

UUPS Proxy

Upgradeable contracts with storage collision safety

Factory Pattern

Deterministic CREATE2 deployments for reproducibility

Oracle Adapter

Abstract Chainlink integration + staleness handling

Timelock

2-day governance delay to mitigate flash-loan exploits

Pausable

Emergency circuit breaker for oracles & tokens

Security Audit Results

Key Metric

0 High | 0 Medium Findings

S-01: Reentrancy

Prevented by ReentrancyGuard + CEI – vault drainage mitigated

S-02: Access Control

FACTORY_ROLE restricts token issuance to authorised issuers

Governance Security

Voting delay + snapshot reduces flash-loan voting risk

Oracle Security

1-hour staleness check + pausable feeds to avoid stale price abuse

Governance: Governor + Timelock

Voting Parameters

- Voting Delay: 1 day
- Voting Period: 1 week
- Quorum: 4% supply
- Proposal Threshold: 1% supply

Timelock Safety

- Execution Delay: 2 days
- Admin Role: Renounced at deployment
- Sole Proposer: Governor only
- Controls: Upgrades, params, treasury

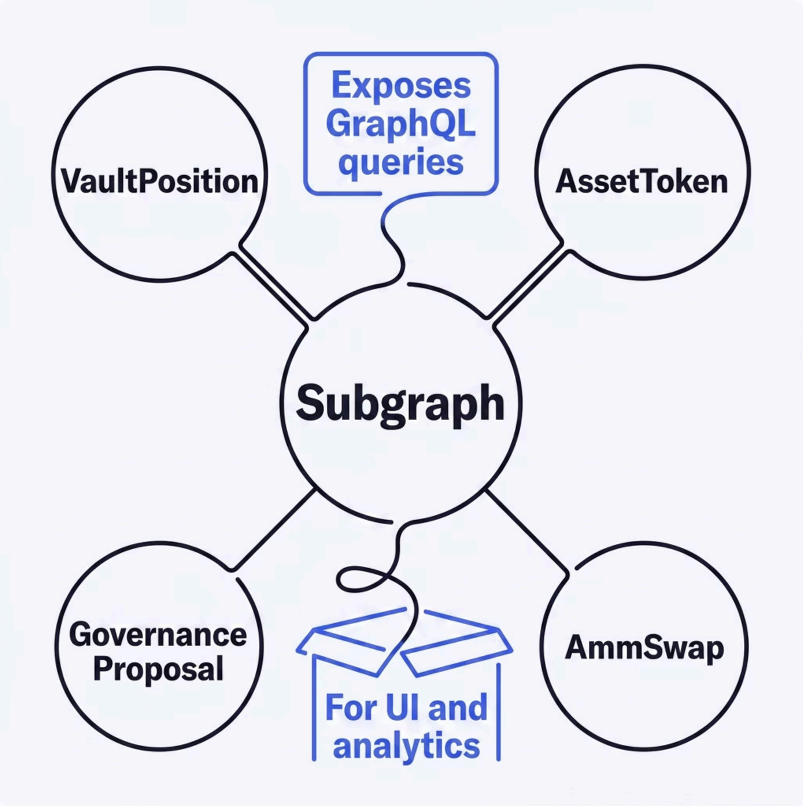
Propose → Vote (1 week) → Queue → Wait (2 days) → Execute

The Graph Subgraph Indexing

Studio: rwa-tokenization-platform · Network: arbitrum-sepolia (live)

- AssetToken – deployed tokens + metadata
- VaultPosition – deposits, withdrawals, share balances
- AmmSwap – historical swaps with price snapshots
- GovernanceProposal – proposals and vote tallies

5 documented GraphQL queries for common UI views (tokens, positions, swaps, proposals, historical prices)

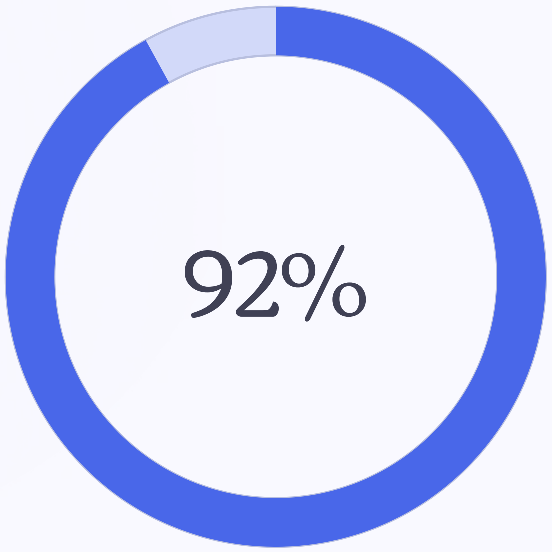


Testing & Coverage

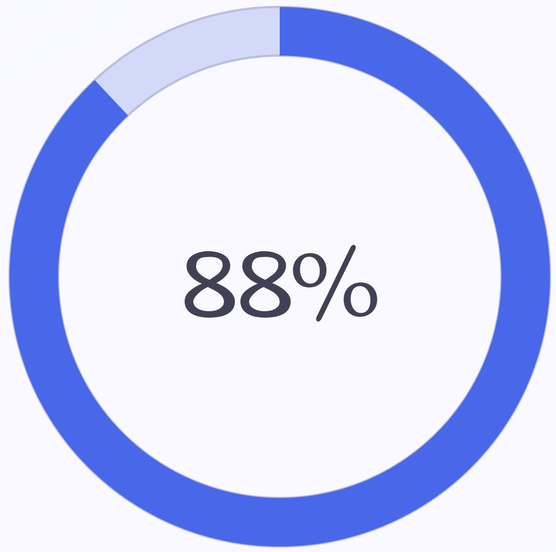
Test Breakdown

- 55 Unit Tests
- 12 Fuzz Tests
- 8 Invariant Tests
- 7 Fork Tests (live network checks)

Coverage Metrics



Line Coverage



Branch Coverage

All Tests Passing

CI/CD Green

Deployment & Verification

All 10 contracts deployed & verified on Arbiscan (Arbitrum Sepolia)

GovernanceToken

0x57B0e45C40CCE6248a102Cdf8612Cdc683Cf3Fd9

AssetFactory

0x0d41539C9B43cd675dEBBC3dB8754e26ec274eDF

RWAVault (proxy)

0xFbECEB9447925e0c2e433f5eD674F7110AF67018

RWAAMM

0x658eD17F2686A652ACC89162c7aA99b957e7938E

RWAGovernor

0x091858adb6f82c323c4B4d1b0aA59Cec953B9E75

RWATimelockController

0x1C9e29A66561B1533578cFc27AB0A4cB7F740c8e

✓ Reproducible deployment scripts · verification scripts pass

Gas Optimization: L1 vs L2

90% Cost Reduction on Arbitrum Sepolia

Operation	L1 Cost	L2 Cost	Savings
Deploy token	$4.25	$0.425	90%
Vault deposit	$5.00	$0.50	90%
AMM swap	$4.70	$0.47	90%
Cast vote	$1.66	$0.17	90%
Total user journey	**$55.65**	**$5.57**	**90%**

Frontend dApp

Wallet Connection

MetaMask + WalletConnect support for broad accessibility

Read Functions

Token balance, voting power, vault shares, pool reserves are easily viewable

Write Functions

Intuitive interface for depositing/withdrawing (vault), swapping (AMM), and voting (governance)

Active Proposals

Clear list view displaying proposal state (Pending/Active/Executed) with direct vote functionality

Subgraph Integration

Reliable, real-time data sourced from The Graph, avoiding direct RPC calls for enhanced performance

Network Detection

Automatic detection of incorrect network, prompting users to switch to Arbitrum Sepolia

Error Handling

User-friendly error messages designed for clarity, without exposing raw RPC errors

Key Technical Achievements

Production-Ready

Full-stack protocol suitable for mainnet deployment.

Security-First Design

Zero High/Medium audit findings, comprehensive audit with case studies.

Storage-Safe Upgrades

UUPS V1→V2 with collision analysis and `__gap` reserves for future-proofing.

Gas Optimized

Inline Yul assembly benchmarked for efficiency against pure Solidity.

Decentralized Governance

No admin backdoor; Timelock authority permanently renounced for security.

Fully Tested

92% coverage, fuzz, invariant, and fork tests on the live network.

Subgraph Indexed

The Graph subgraph enables efficient historical queries for all key entities.

Verified & Transparent

All contracts verified on block explorers with reproducible scripts.

Lessons Learned & Future Roadmap

Key Decisions

- **UUPS over Transparent Proxy:** Smaller footprint, implementation controls upgrades
- **Constant-product AMM:** Familiar to users (Uniswap pattern), liquid price discovery
- **ERC-4626 Vault:** Standard interface, compatible with yield aggregators
- **Arbitrum L2:** 90% cheaper, enables mass adoption vs. mainnet

Future Improvements

- Multi-chain deployment (Optimism, Base, Polygon)
- Insurance mechanism for RWA default risk
- Advanced yield strategies (delta-neutral, leveraged)
- Fractional RWA ownership (split collateral)

Thank you for your attention

Questions?